

Design-Decision Memo: Fari Sensor Hub

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Project: Week 5 Mini-Capstone (Fari Sensor Hub)

1. Overview

This memo documents the engineering rationale behind the hardware design for the Fari-class eldercare assistant sensor hub. The design focuses on integrating an ESP32-S3 module, a multi-modal sensor stack, and a robust power management subsystem within a compact 40 × 40 mm footprint. The objective is to provide a reliable, low-noise perception interface for mobile robotic applications, ensuring functional safety and electromagnetic compatibility in an indoor, human-interactive environment.

2. Key Design Decisions

2.1 Power Distribution & Safety

The power architecture centers on the AP2112K-3.3 low-dropout (LDO) regulator. This component was selected for its high Power Supply Rejection Ratio (PSRR) and low quiescent current, which are essential for maintaining the clean, stable voltage rails required by precision I2C sensors. To ensure safety and reliability in an eldercare setting, the input path is protected by a polyfuse (MF-MSMF050-2) and a TPD4E05U06DQA ESD protection array. The choice of a self-resetting polyfuse over a traditional one-time fuse reflects a strategic decision to minimize manual maintenance requirements, while the TVS diode array provides robust protection against electrostatic discharge during frequent human-robot interactions.

2.2 Sensor Integration and Communication

To manage the multi-sensor stack effectively, I integrated a PCA9548A I2C bus expander. This decision was driven by the need to resolve potential I2C address conflicts among the IMU, Time-of-Flight (ToF) ranger, and environmental sensors, while simultaneously providing a modular architecture for future hardware expansion. Signal integrity was maintained through meticulous bus capacitance management; by keeping the estimated total bus capacitance well below the 400 pF threshold, we ensure reliable data transmission at standard 400 kHz I2C speeds. Furthermore, localized 0.1 μF decoupling capacitors were positioned immediately adjacent to every IC's VCC pin to minimize local supply impedance and suppress high-frequency noise.

2.3 Layout and RF Performance

RF performance remains a critical constraint for the ESP32-S3 module. Consequently, a strict Keep-out Zone was implemented directly beneath the antenna trace to prevent any copper pours, traces, or vias from interfering with the antenna's radiation pattern. This was enforced by modifying the component footprint properties to ensure that no conductive materials encroach upon this critical area. Additionally, the IMU was placed near the board's geometric center and rotated at a 45-degree angle. This placement strategy is intended to minimize the impact of mechanical vibrations on sensor data, which is a common point of failure in mobile robotics.

3. Engineering Trade-offs and Considerations

Several design trade-offs were evaluated during the development of this hub. First, a 2-layer PCB stack-up was selected to balance manufacturing cost against the required routing density; this configuration successfully satisfied all target complexity requirements while remaining compliant with standard 6/6 mil manufacturing design rules. Second, although the integration of an I2C expander increases the overall Bill of Materials (BOM) cost and component count, the trade-off in favour of increased modularity and simplified address management was deemed necessary for the long-term flexibility of the Fari platform. Finally, the decision to utilize a polyfuse, despite the slight increase in series resistance compared to a standard trace or zero-ohm link, was non-negotiable to satisfy the safety and reliability standards inherent to consumer-facing assistive robotics.

4. Conclusion

The current design fulfills all functional, electrical, and mechanical specifications established in the Week 5 project requirements. By adhering to these rigorous design choices and successfully clearing both ERC and DRC checks, the sensor hub is now fully validated for the final stage of manufacturing output generation, including the production of Gerbers, drill files, and pick-and-place data.